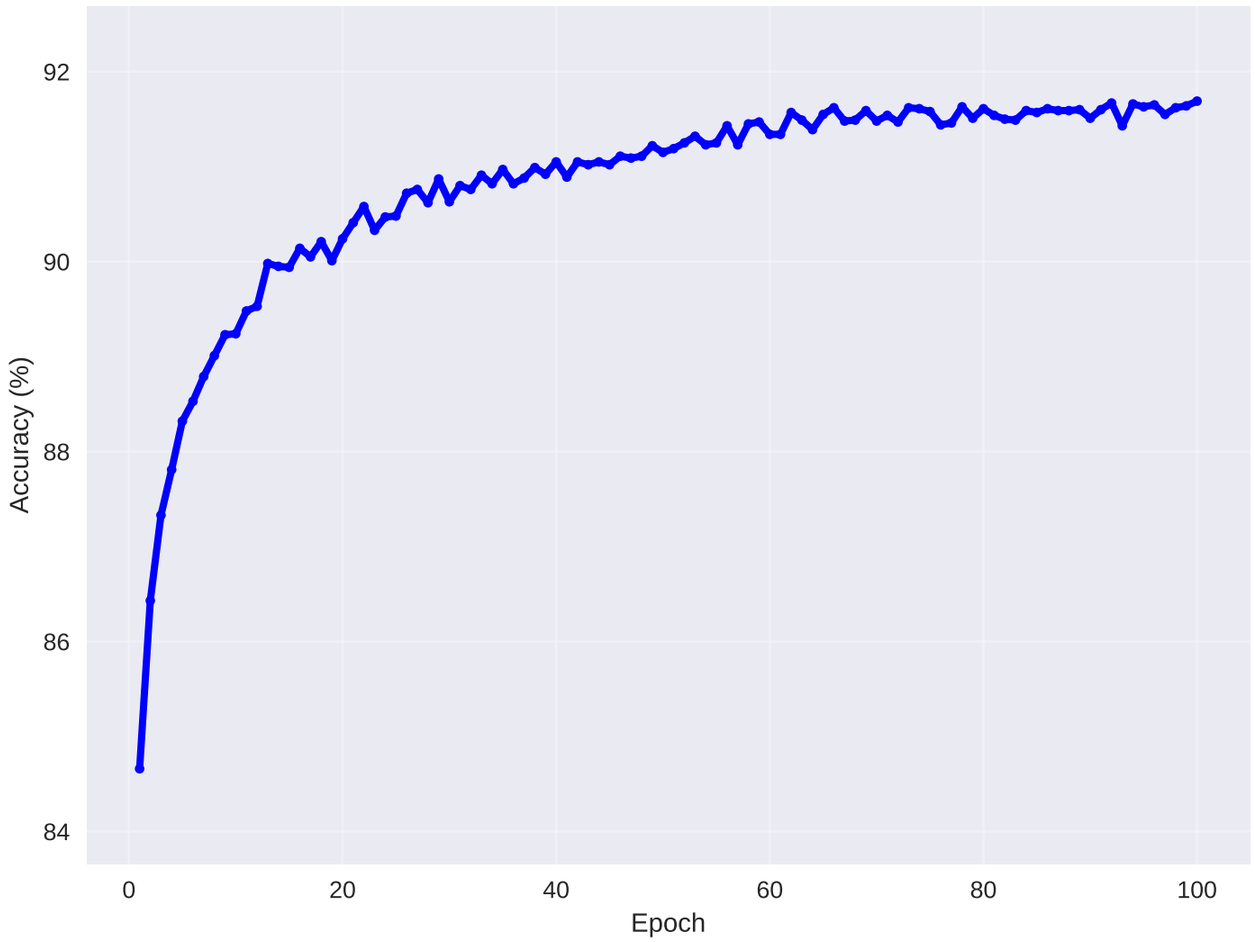
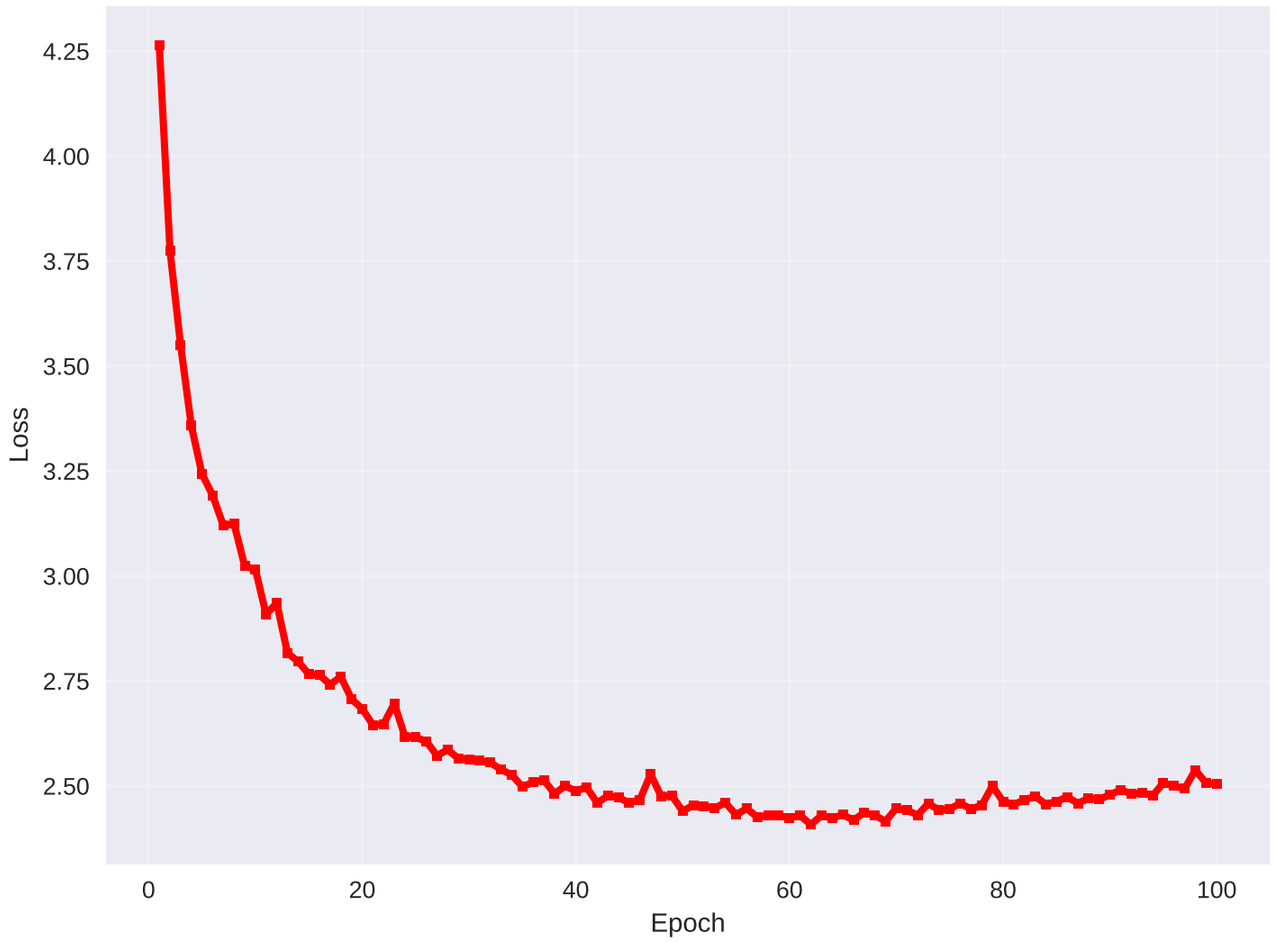
Federated Learning - Main Metrics Overview

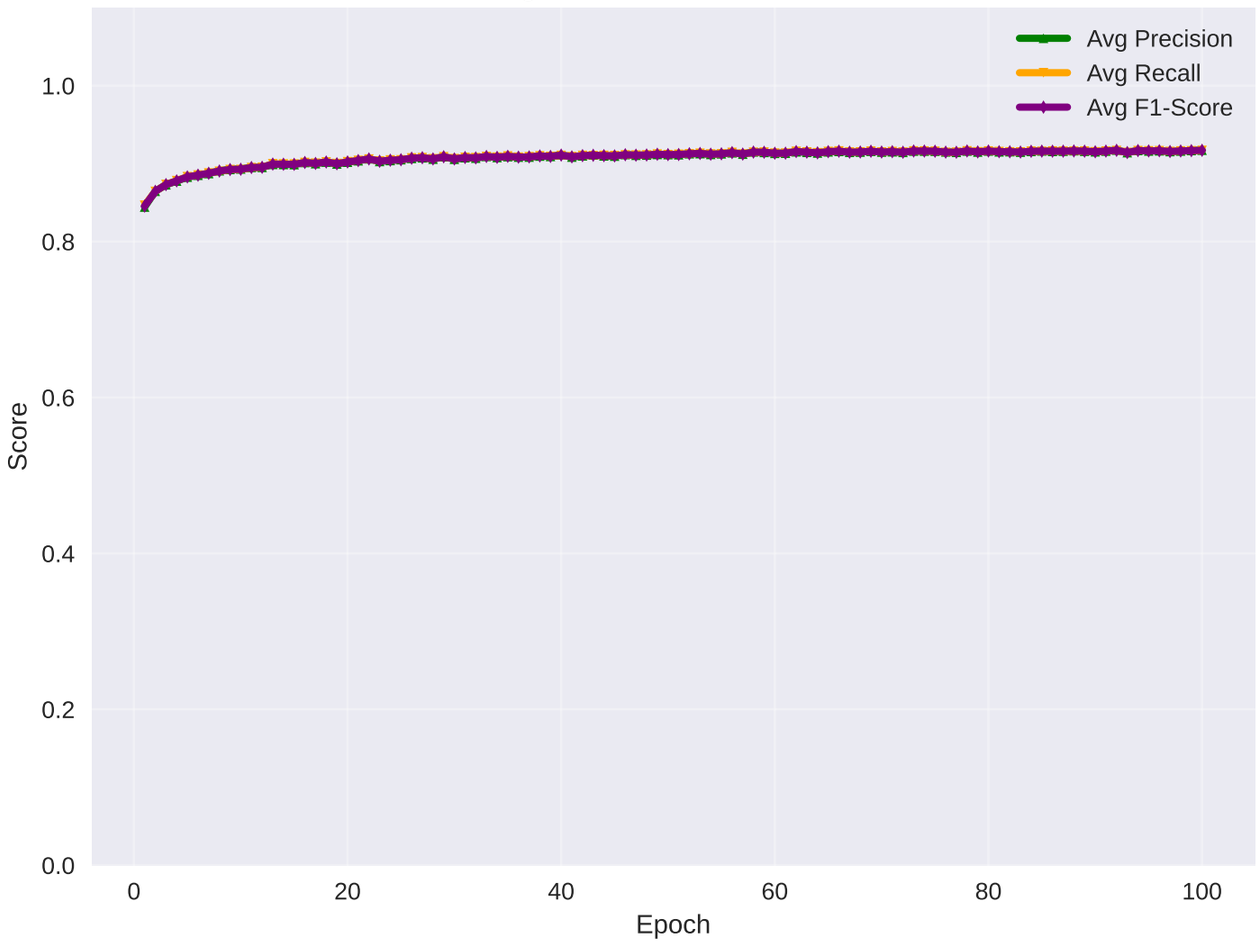
 Global Model Accuracy Evolution



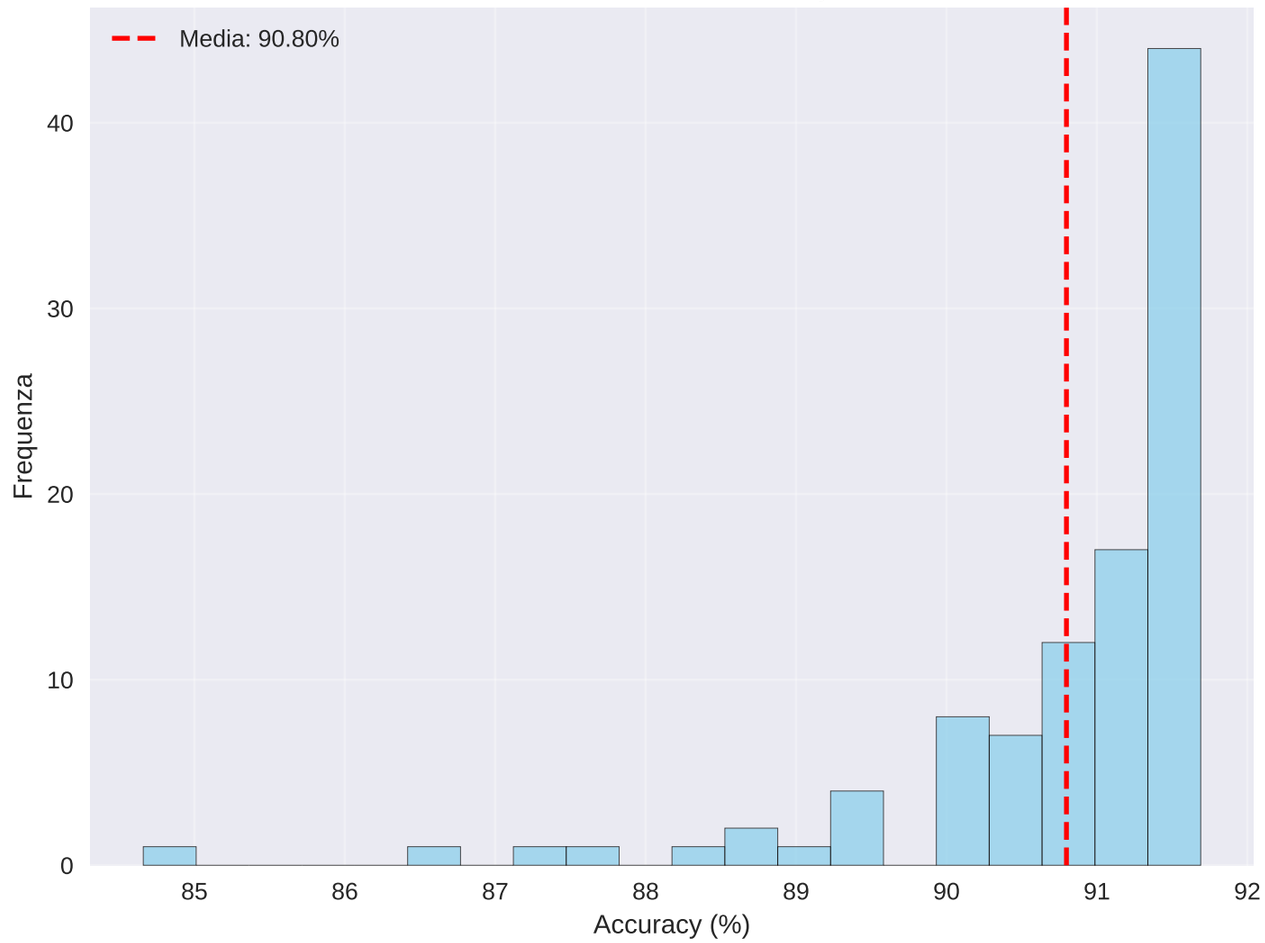
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

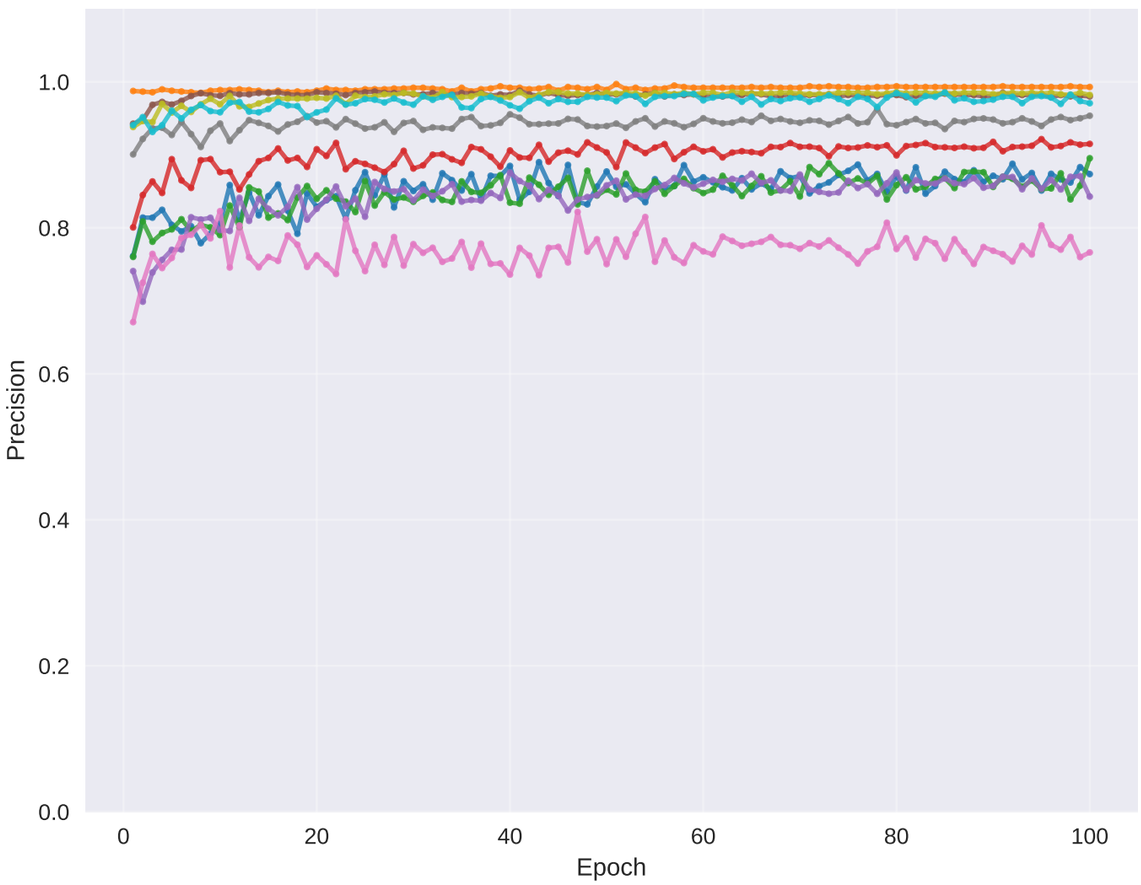


 Distribuzione Accuracy

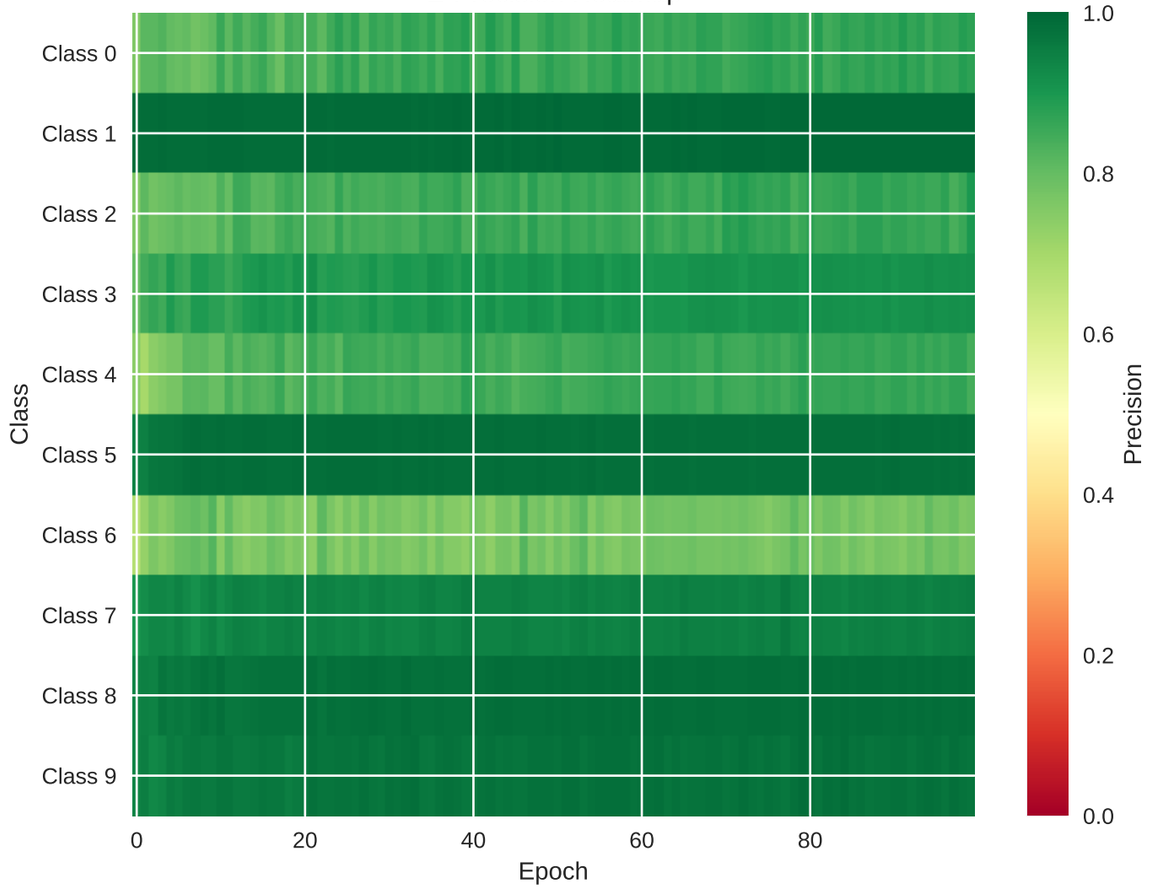


Per-Class Precision Analysis

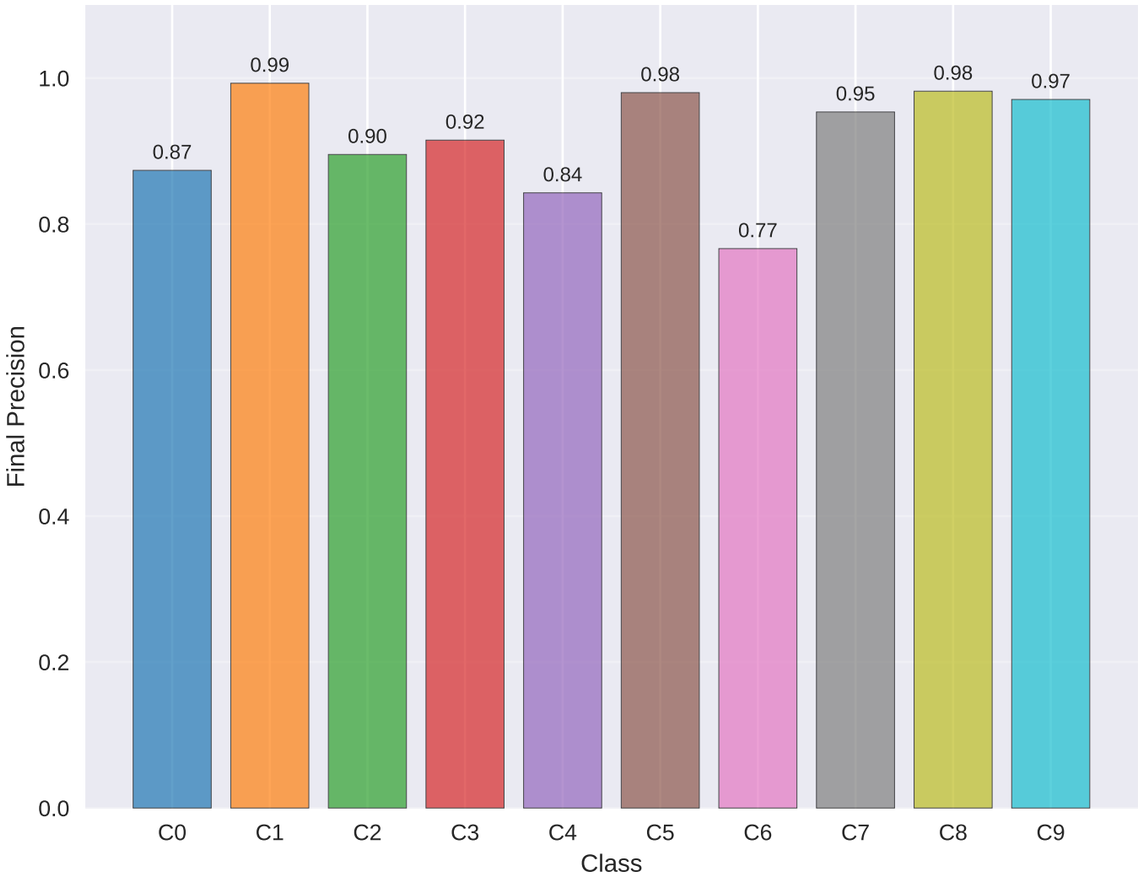
□ Per-Class Precision Evolution



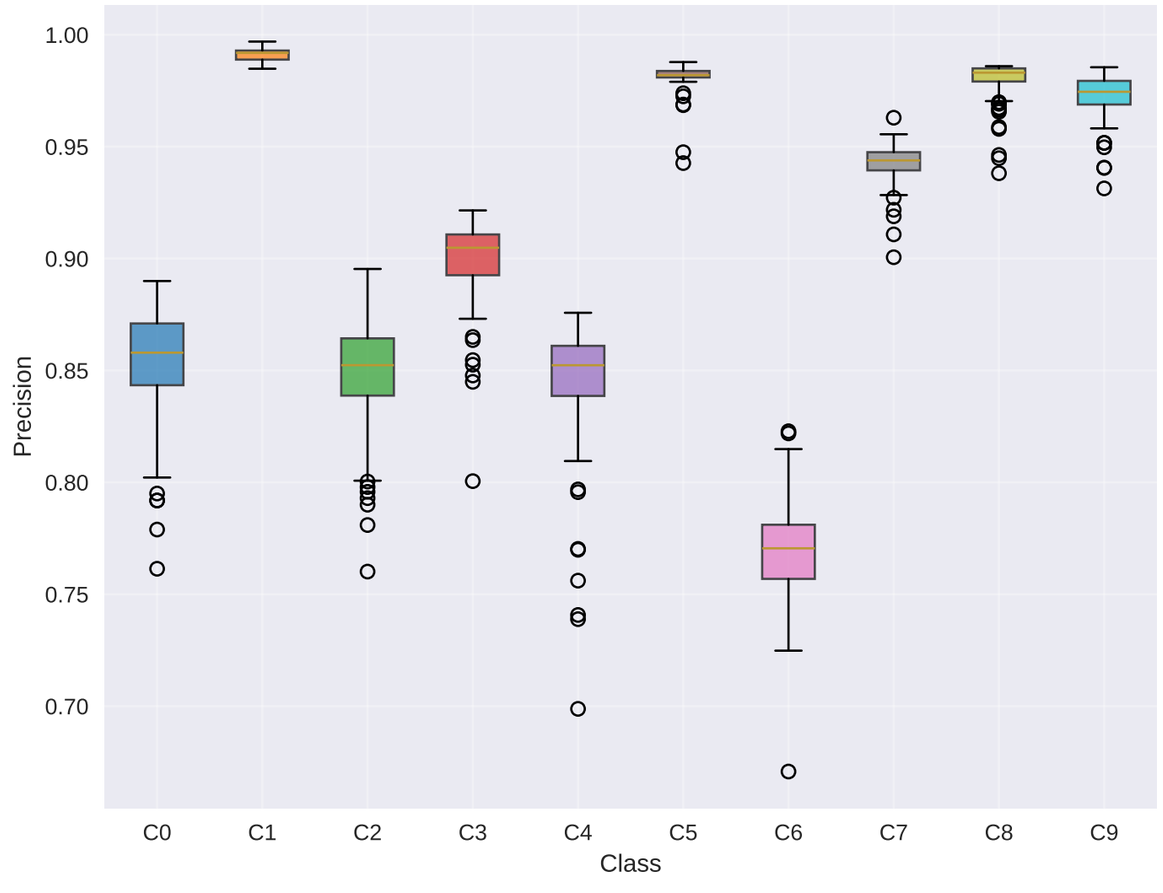
□ Precision Heatmap



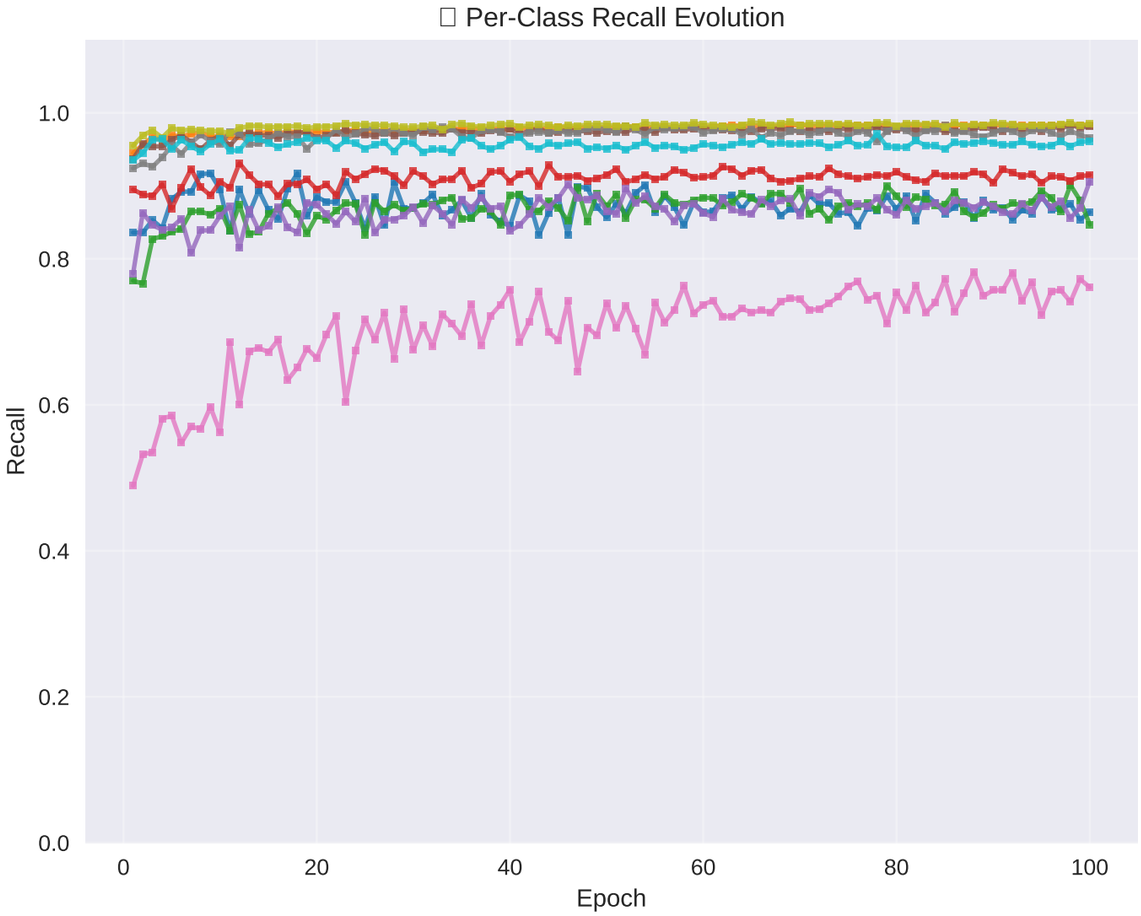
□ Final Precision per Class

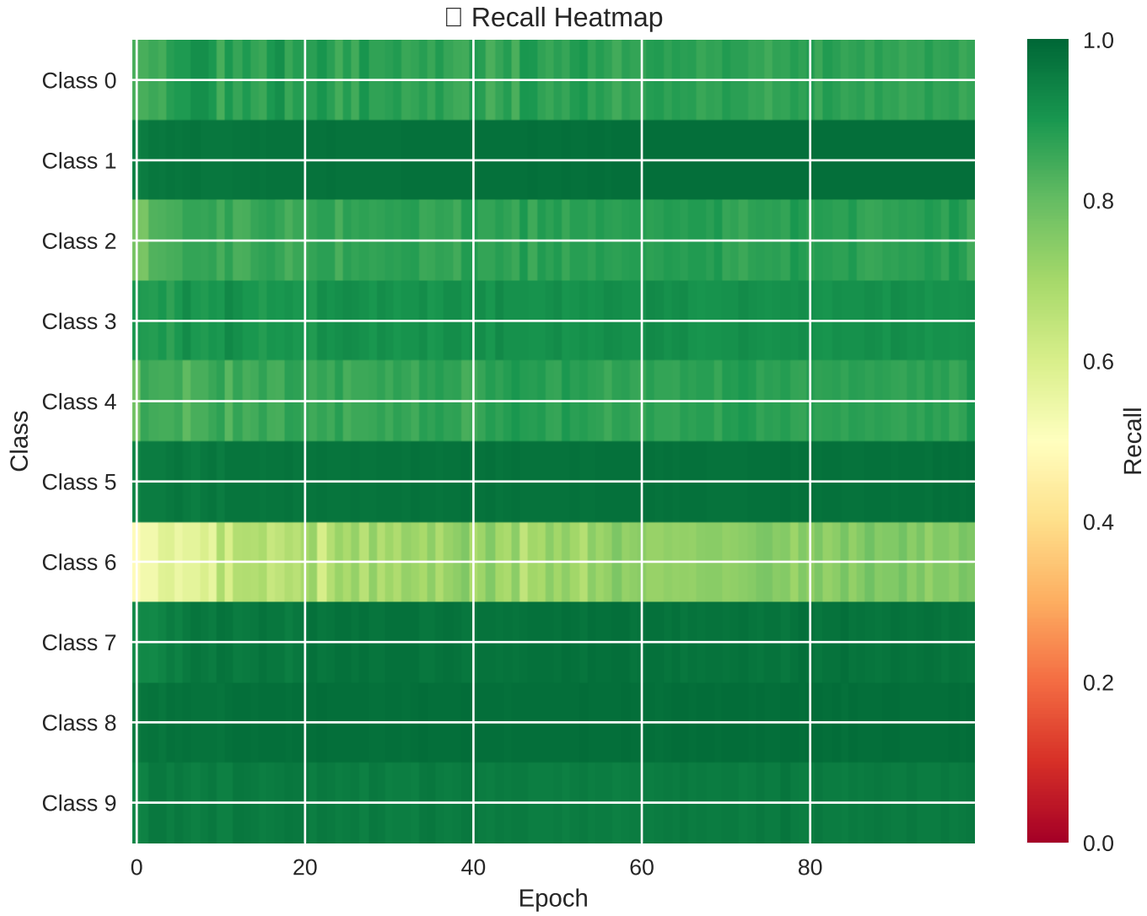


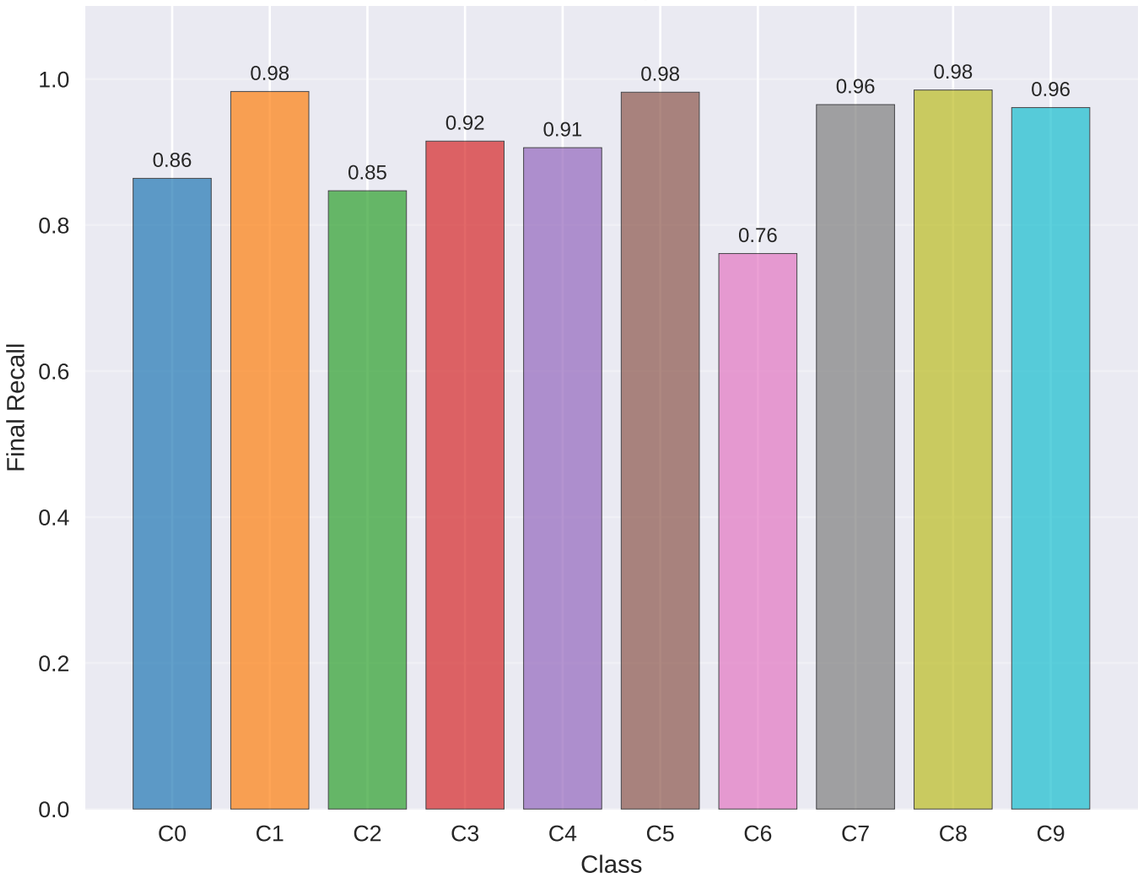
□ Precision Distribution per Class

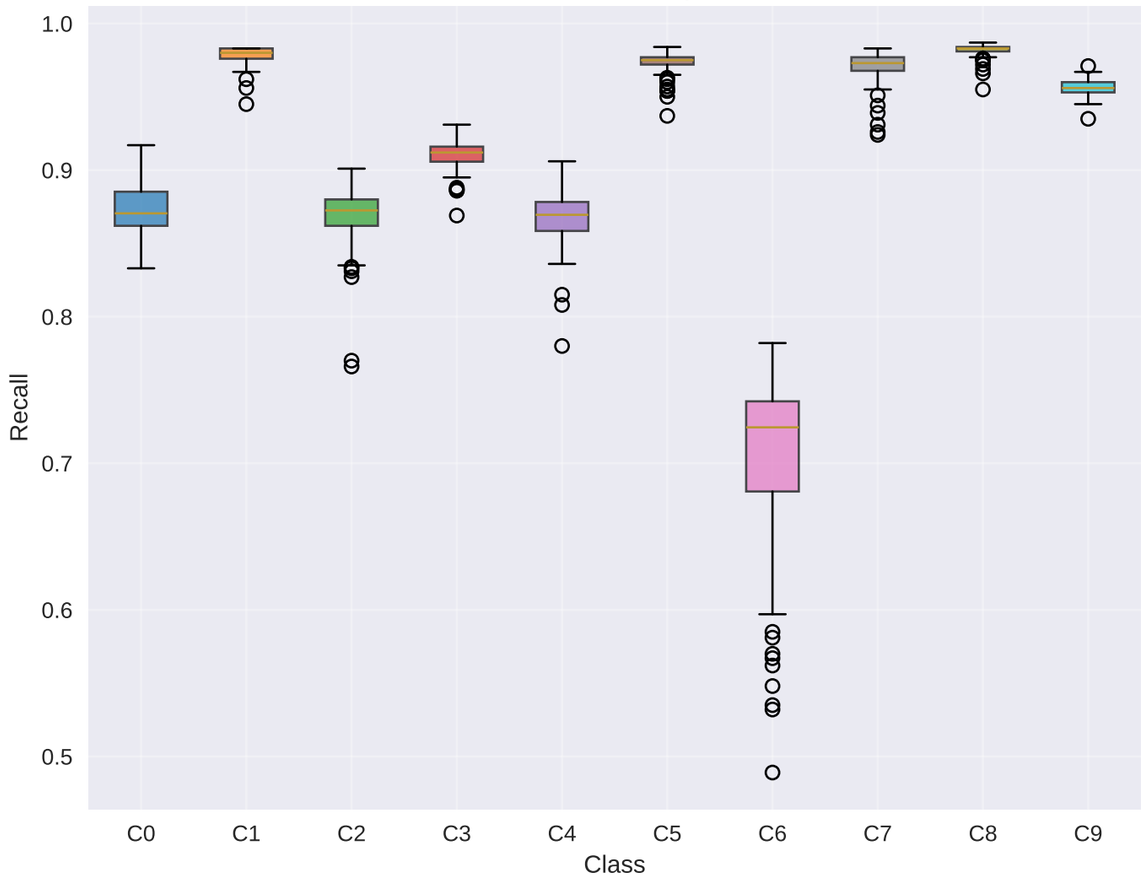


Per-Class Recall Analysis

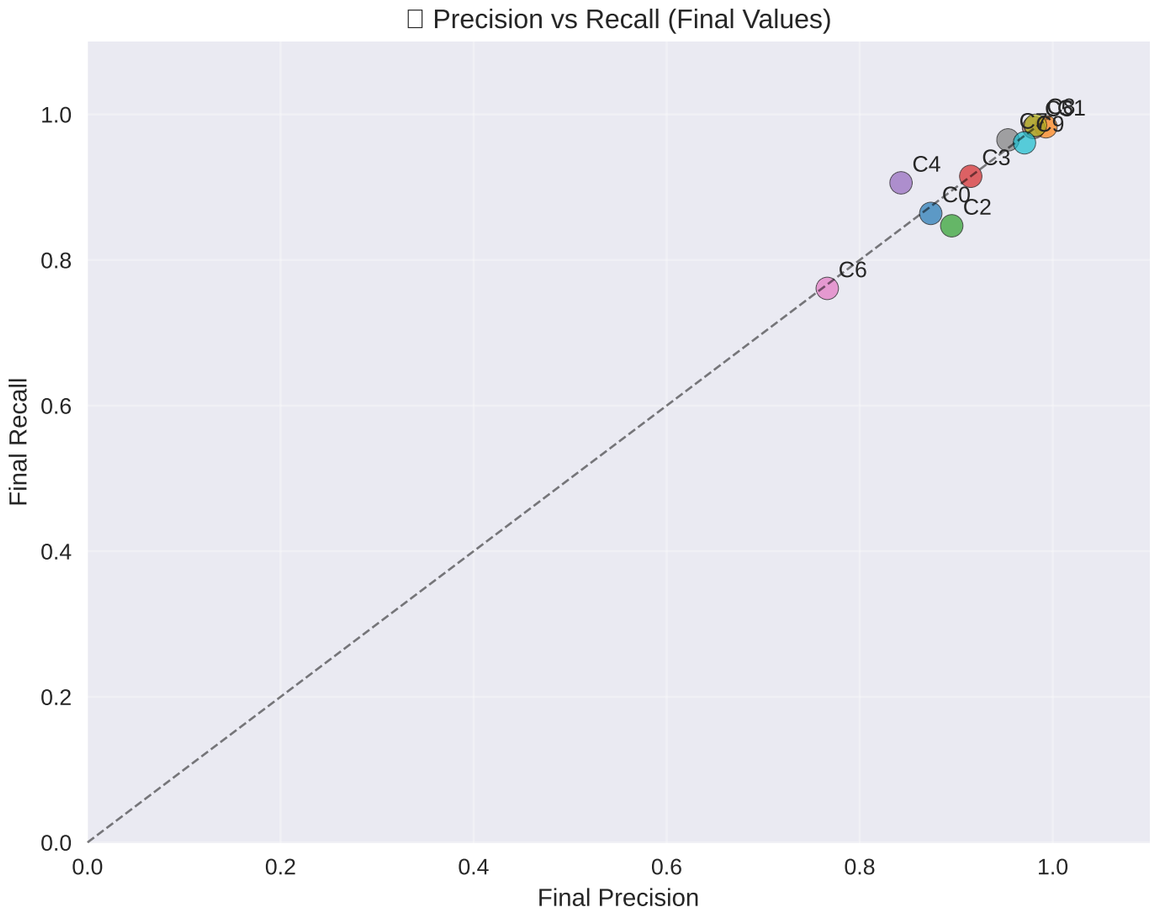


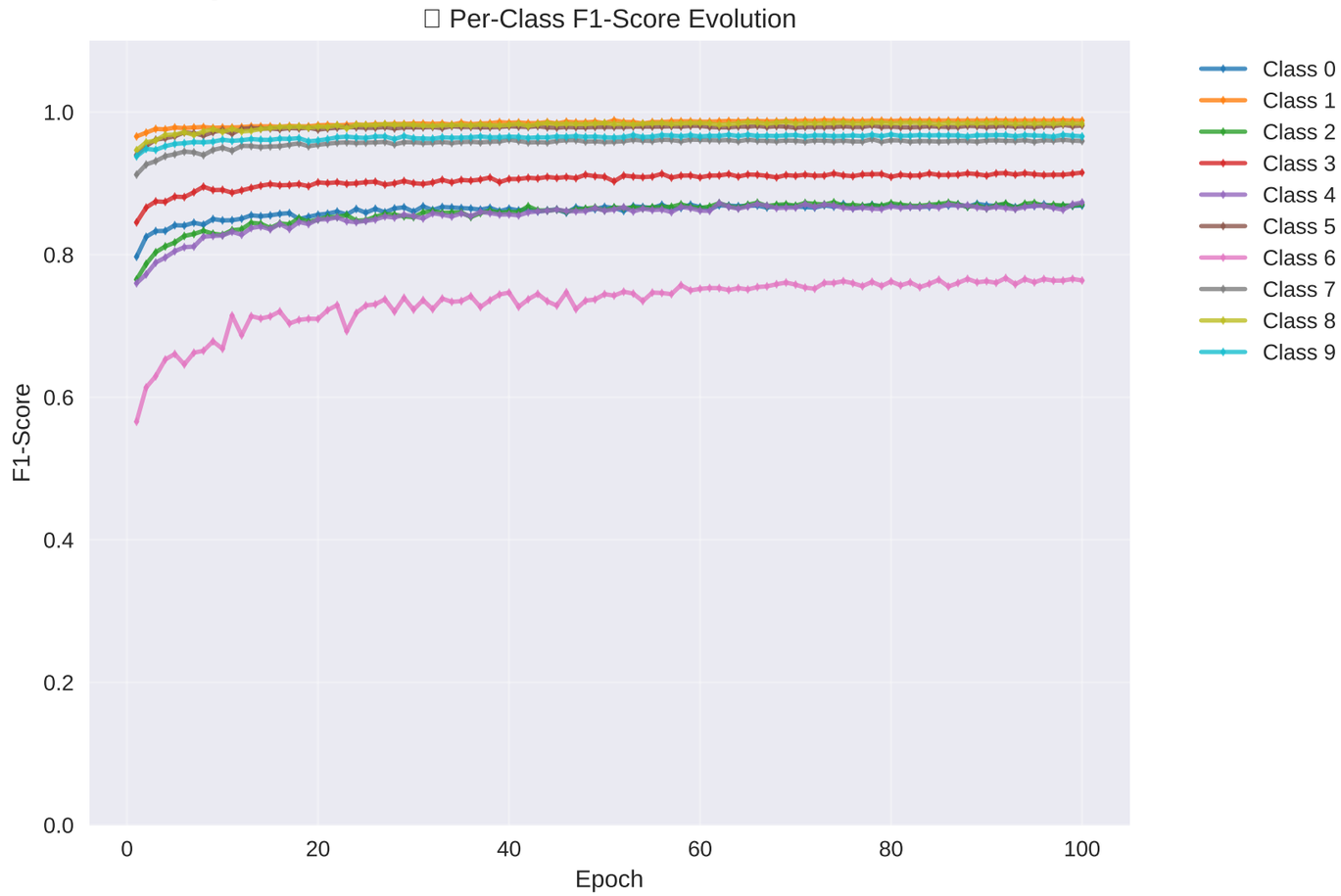


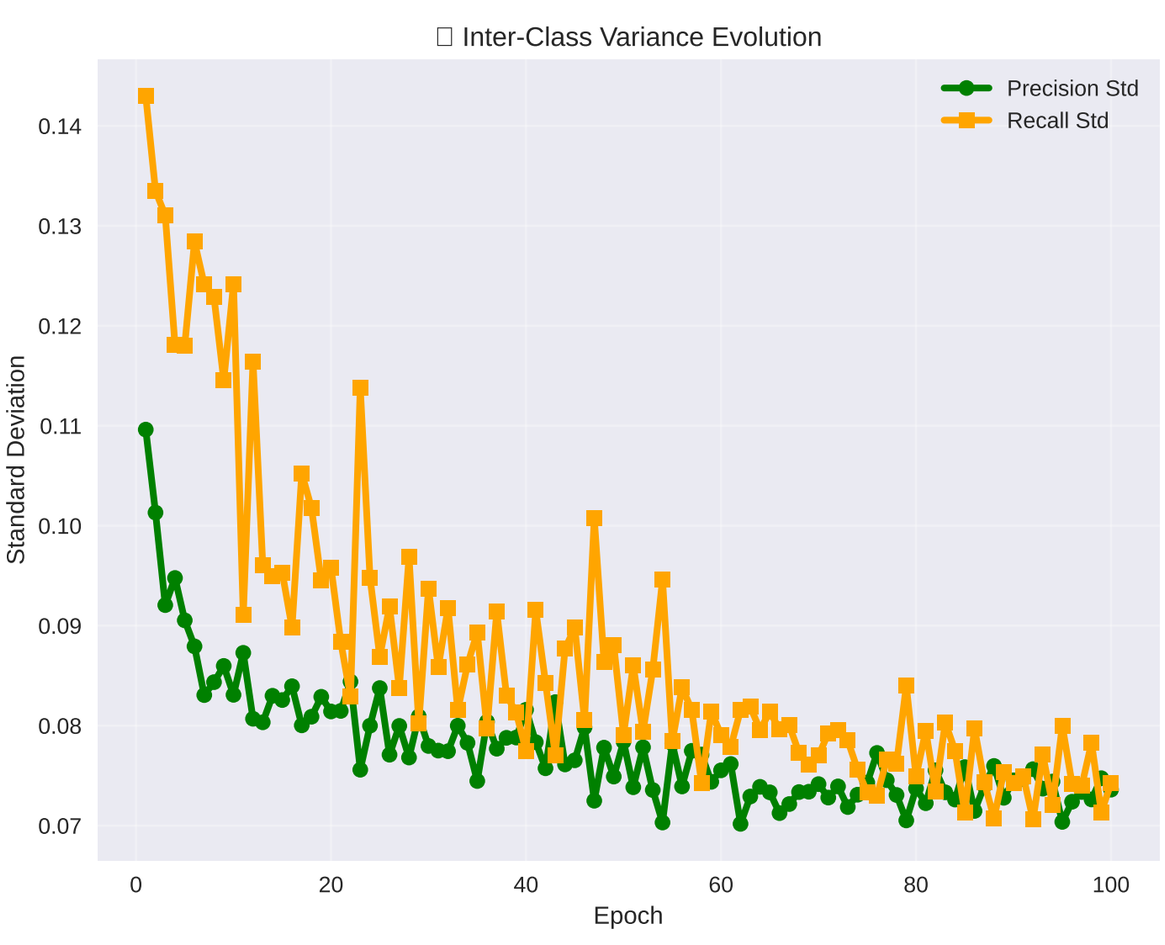


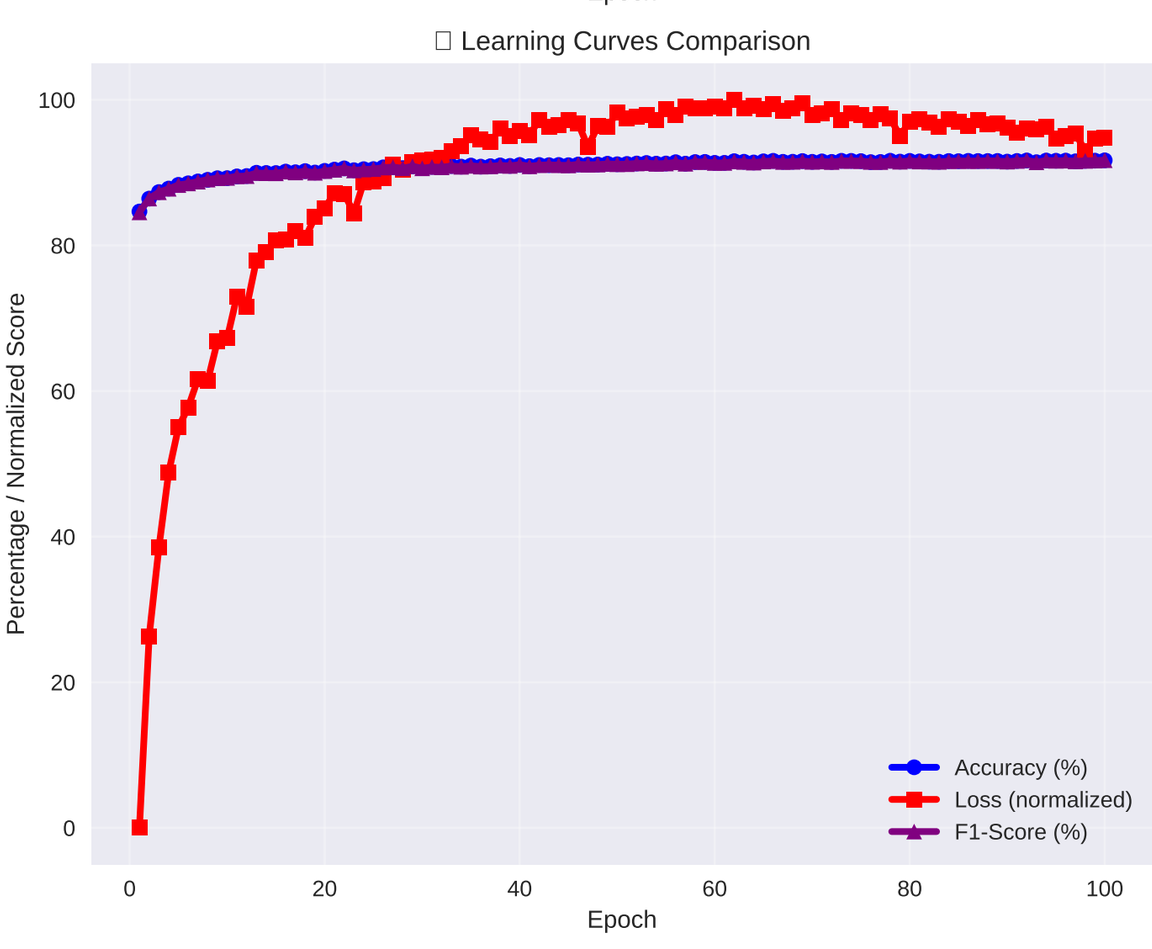


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-01 13:46:31
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.69%
• Loss: 2.5042
• Average Precision: 0.917
• Average Recall: 0.917
• Average F1-Score: 0.917

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +7.03%
• Riduzione Loss: +1.7601
• Stabilità Accuracy (CV): 0.0129

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.766)
• Migliore Recall: Classe 8 (0.985)
• Peggiore Recall: Classe 6 (0.761)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.874, R=0.864, F1=0.869
• Classe 1: P=0.993, R=0.983, F1=0.988
• Classe 2: P=0.895, R=0.847, F1=0.871
• Classe 3: P=0.915, R=0.915, F1=0.915
• Classe 4: P=0.843, R=0.906, F1=0.873
• Classe 5: P=0.980, R=0.982, F1=0.981
• Classe 6: P=0.766, R=0.761, F1=0.764
• Classe 7: P=0.954, R=0.965, F1=0.959
• Classe 8: P=0.982, R=0.985, F1=0.984
• Classe 9: P=0.971, R=0.961, F1=0.966

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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